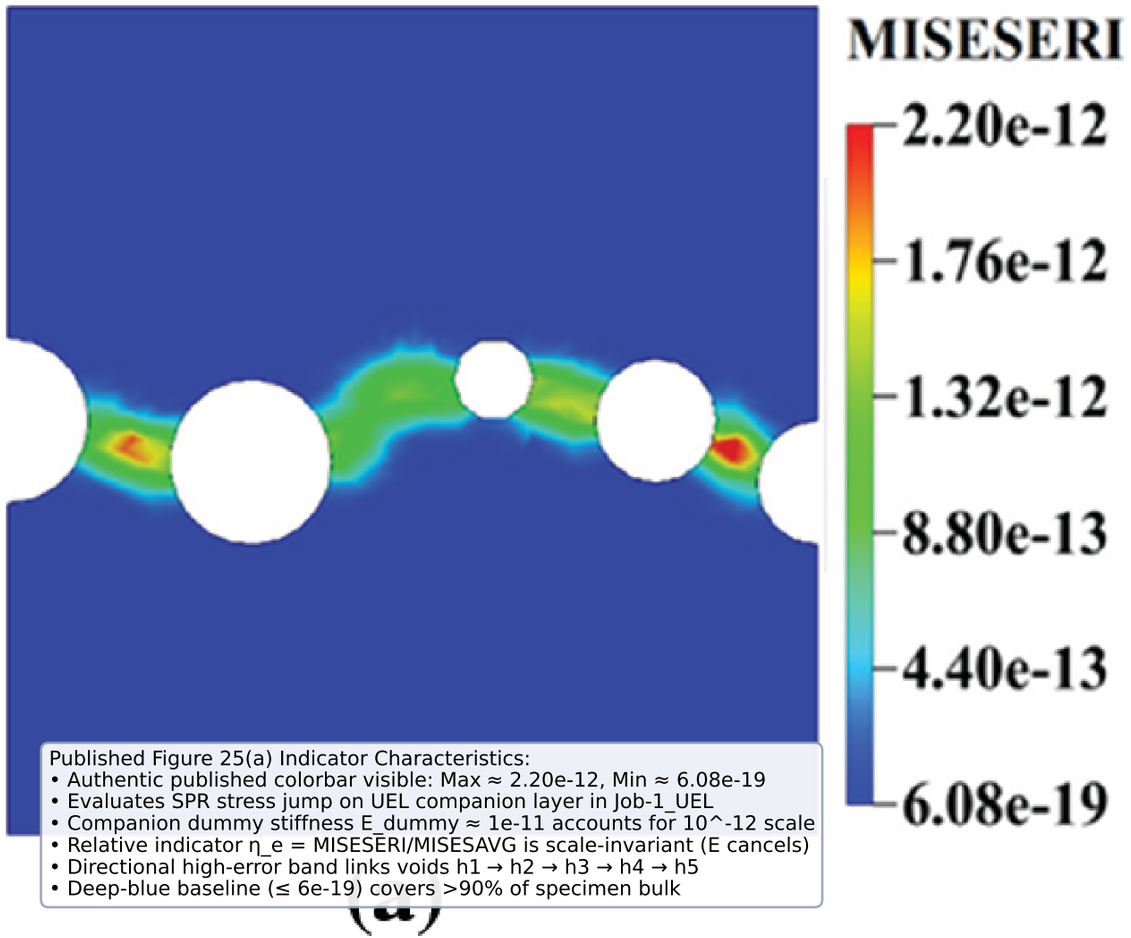
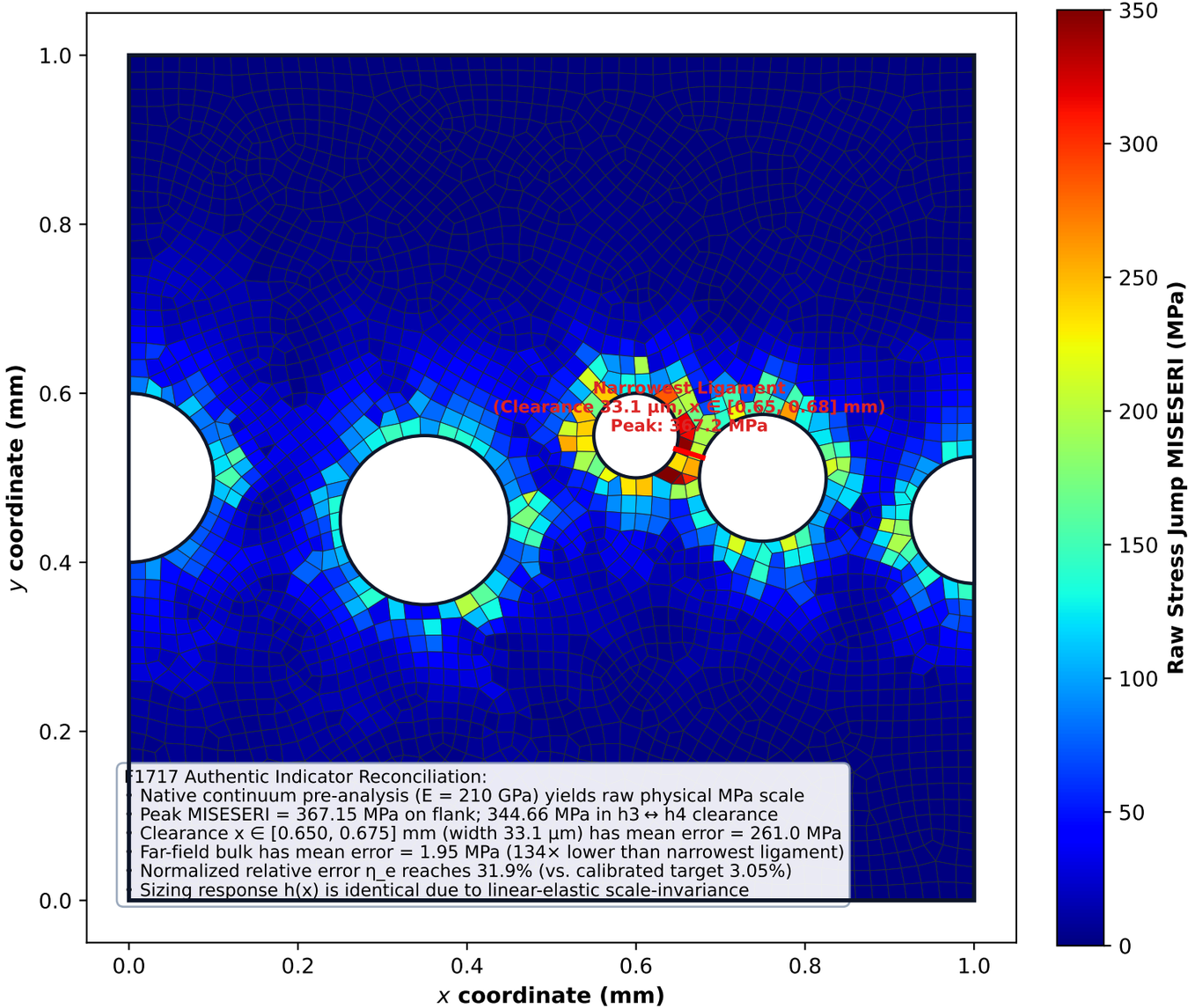
(a) Published Five-Hole Error Indicator (Fig. 25a with Authentic Legend)
(Pandey & Kumar 2025, Section 4.5, Page 3279, xref=803)



Published Figure 25(a) Indicator Characteristics:
- Authentic published colorbar visible: Max ≈ 2.20e-12, Min ≈ 6.08e-19
- Evaluates SPR stress jump on UEL companion layer in Job-1_UEL
- Companion dummy stiffness E_dummy ≈ 1e-11 accounts for 10^-12 scale
- Relative indicator η_e = MISESERI/MISESAVG is scale-invariant (E cancels)
- Directional high-error band links voids h1 → h2 → h3 → h4 → h5
- Deep-blue baseline (≤ 6e-19) covers >90% of specimen bulk

(b) Authentic Raw MISESERI (Job-1 Coarse Pre-Analysis)
(Abaqus SPR Mises Stress Jump Indicator, 2,747 Physical Elements)



F1717 Authentic Indicator Reconciliation:
- Native continuum pre-analysis (E = 210 GPa) yields raw physical MPa scale
- Peak MISESERI = 367.15 MPa on flank; 344.66 MPa in h3 ↔ h4 clearance
- Clearance x ∈ [0.650, 0.675] mm (width 33.1 μm) has mean error = 261.0 MPa
- Far-field bulk has mean error = 1.95 MPa (134× lower than narrowest ligament)
- Normalized relative error η_e reaches 31.9% (vs. calibrated target 3.05%)
- Sizing response h(x) is identical due to linear-elastic scale-invariance